Rachit Revalia

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Portfolio

Professional Summary

Python Software Engineer and **Data Engineer** with experience building production-grade **data pipelines**, **computer vision** systems, and **FastAPI** backends. Experienced in **Python**, **PostgreSQL**, **machine learning**, and cloud-connected applications, with practical industry projects in esports analytics and AI. Currently completing a B.Eng. in ICT at JAMK University of Applied Sciences, specializing in **Data Analysis and Artificial Intelligence**, graduating December 2026, with a two-semester exchange in Applied Data Science & AI at BUas (Netherlands).

Technical Skills

Languages: Python (advanced), TypeScript, JavaScript, SQL, Bash/Shell, C++ (basic)
Backend & APIs: FastAPI, Flask, Django, PostgreSQL, SQLite, REST APIs, JWT Auth
Frontend: React, TypeScript, Vite, Electron
Data & ML: Pandas, NumPy, scikit-learn, PyTorch, CUDA, YOLOv11, Grad-CAM, OpenCV, Plotly
AI & Agentic Systems: LLM APIs (Gemini), Playwright, Pydantic, agentic automation, Claude Code
Infrastructure & Tools: Git, Linux/Bash, Docker, SSH/SFTP, Paramiko

Experience

Experience Coordinator (Volunteer) **Jul. 2026 – Present**
Spinout, Jyväskylä Entrepreneurship Society *Jyväskylä, Finland*

- Selected from 300+ applicants to join the founding team of a new deeptech initiative connecting university research with startup formation; design the participant experience for events and workshops.

Data Scientist (Industry Project) **Feb. 2026 – Jun. 2026**
PlaySmart × Breda Guardians Esports, BUas *Breda, Netherlands*

- Embedded with a professional **esports** coaching team to understand their operational problems, then built and deployed production solutions used daily, owning the entire arc from first conversation to shipped software.
- Built a **YOLOv11** GPU-accelerated **computer vision** pipeline (**CUDA**/**PyTorch**) with **Grad-CAM explainability** to automate reaction time analysis, cutting processing from 2–3 hours to 42 minutes per session and surfacing model decisions clearly to non-technical coaching staff.
- Designed and shipped a **FastAPI** backend with server-side pandas analytics, replacing 100+ batched frontend HTTP calls with efficient single-query computation, a decision made under real constraints with real performance consequences for end users.
- Integrated Nuanic **BLE** wearable sensors for real-time biometric capture; diagnosed a MAC address rotation bug in the Windows BLE stack causing silent data corruption by digging into root cause, not patching around symptoms.

Data Engineer (Industry Project) **Sep. 2025 – Jan. 2026**
B.G.E.T. × Breda Guardians Esports, BUas *Breda, Netherlands*

- Owned a greenfield project end-to-end: designed **ETL pipelines** processing thousands of gameplay and biometric events, integrating Riot Games, Henrik's API, and Tobii Eye Tracker data into **PostgreSQL** analytics workflows, and shipped a **React**/**TypeScript Electron** desktop app, all deployed to production on a Linux server within a single semester.
- Wrestled with messy, heterogeneous data from sources with different sampling rates and no shared timestamp anchor; built a matching engine to align them reliably, with validation logic to catch data quality issues before they propagated downstream.
- Worked directly with coaching staff to understand what mattered, iterated fast on real user feedback, and made architecture decisions with incomplete information rather than waiting for perfect requirements.

AI Code Trainer & Evaluator **Nov. 2024 – Dec. 2024**
Outlier · Remote Internship *Finland*

- Evaluated **AI**-generated **Python** code for correctness and efficiency; developed structured rubrics and comprehensive test suites, exercising independent judgment on quality and edge cases at scale.

Projects

Computer-Use Automation System | *Python, Playwright, Flask, Pydantic, Gemini API* | GitHub **2026**

- Built an **LLM**-driven web automation **agent** where the model discovers how to complete tasks in real browser UIs with no API access; successful runs are captured as typed, versioned, reusable capability artifacts that replay with zero **LLM** calls, combining agentic AI with deterministic execution.

Task Manager | *Python, Django, PostgreSQL, JavaScript, Bootstrap* | GitHub **2024**

- Full-stack web application with secure authentication, task assignment, and deadline tracking; end-to-end ownership from schema design to deployed service.

Library Management System | *Python, SQLite, CLI* | GitHub **2023**

- Cross-platform Python CLI application with full CRUD operations, input validation, and zero external dependencies.

Education

JAMK University of Applied Sciences **Aug. 2023 – Dec. 2026**
B.Eng. in ICT — Data Analysis & AI (Graduating Dec. 2026) *Jyväskylä, Finland*

Breda University of Applied Sciences (BUas) **Sep. 2025 – Jun. 2026**
Exchange Year– Applied Data Science & Artificial Intelligence *Breda, Netherlands*

Languages

Hindi: Native **English**: Professional Working Proficiency (CEFR: C1 overall) **Finnish**: A2